6.5.3 Using ultrasound

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	ultrasound; longitudinal wave with frequency greater than 20 kHz	
(b)	piezoelectric effect; ultrasound transducer as a device that emits and receives ultrasound	
(c)	ultrasound A-scan and B-scan	
(d)	acoustic impedance of a medium; $Z = \rho c$	
(e)	reflection of ultrasound at a boundary;	*M0.3*

(g) Doppler effect in ultrasound; speed of blood in the patient; $\frac{\Delta f}{f} = \frac{2v\cos\theta}{c}$ for determining the speed v of blood.

2d. Prior learning and progression

This specification has been developed for learners who wish to continue with a study of physics at Level 3. The A level specification has been written to provide progression from GCSE Science, GCSE Additional Science, GCSE Further Additional Science, GCSE Physics or from AS Level Physics; achievement at a minimum of grade C (or equivalent) in these qualifications should be seen as the normal requisite for entry to A Level Physics. However, learners who have successfully taken other Level 2 qualifications in Science or Applied Science with appropriate physics content may also have acquired sufficient knowledge and understanding to begin the A Level Physics course.

There is no formal requirement for prior knowledge of physics for entry onto this qualification. Other learners without formal qualifications may have acquired sufficient knowledge of physics to enable progression onto the course.

Some learners may wish to follow a physics course for only one year as an AS, in order to broaden their curriculum, and to develop their interest and understanding of different areas of the subject. Others may follow a co-teachable route, completing the one–year AS course and/or then moving to the two–year A level developing a deeper knowledge and understanding of physics and its applications.

The A Level Physics course will prepare learners for progression to undergraduate study, enabling them to enter a range of academic and vocational careers in mathematics-related courses, physical sciences, engineering, medicine, computing and related sectors. For learners wishing to follow an apprenticeship route or those seeking direct entry into physical science careers, this A level provides a strong background and progression pathway.

There are a number of Science specifications at OCR. Find out more at www.ocr.org.uk.